

MEC 427 Advanced Sensors and Actuators Challenges 1-5

Date & Time

- **Report Date:** 11-08-2025
 - **Time Block:** 11-03 – 11-08
 - **Time Spent This Session:** 7 hours / minutes
 - **Cumulative Time on Project:** 21 hours / 30 minutes
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Project Overview

- **Project Title:** Challenges 1-3: Smart Sensor and Actuator System
 - **Project Objective:** Create a smart fluid control system using level sensors and a servo-actuated pinch valve to control flow.
 - **Current Phase:** Implementation Phase
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Task / Activity Details

- **Module / Subsystem:** Voting sensor system.
 - **Specific Task or Activity:**
 - **Time Spent on This Task:** 7 hours / minutes
 - **% of Total Time:** 100%
 - **Resources Used:**
 - **Hardware:** Adafruit Circuit Playground Express (CPX), TowerPro SG92R servo, alligator clips, 1/4" aquarium tubing, 3D printer (PLA filament)
 - **Software:** CircuitPython, VSCode IDE
 - **Tools:** Copilot/Claude AI, Internet Research, Python.
 - **Documentation:** Code files.
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Progress Tracking

- **Task/Module:** Challenge 1
- **Project Completion Status:** 100% complete
- **Milestones Reached:** Completed fluid level sensor with capacitive touch.
- **Blockers or Issues:** Learning about calibration.
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Task/Module: Challenge 2

- **Project Completion Status:** 80% complete
- **Milestones Reached:** Sensor array prototype completed.
- **Blockers or Issues:** Need to adapt the tower to hold multiple sensors securely.

Results & Deliverables

- **What Was Achieved:** Completed Challenge 1 and made significant progress on Challenge 2.
 - **Artifacts Created:** CircuitPython code for Challenges 1 and 2, sensor array prototype.
 - **Validation / Testing Outcomes:** Sensor can detect water level
 - **Changes Made to Design or Approach:**
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Narrative & Observations

Use this section for detailed descriptions, reflections, and context. Include technical notes, emotional observations, and rationale for decisions.

- **Narrative Summary:** I made good progress on the software for the sensor and actuator system. In challenge 1 I added a calibration UI to help set the capacitive threshold for water detection. I found I need to adapt the tower to hold multiple sensors securely for challenge 2. Alligator clips are proving to be difficult. I also need a new work area as my desk dosen't have enough space for the full setup. and my cables are too short to arrange everything properly.
 - **Technical Notes:**
 - **Lessons Learned / Insights:**
 - **Emotional/Workflow Reflections (optional):**
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